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8 authors, including:

Salequzzaman Khan
BRAC University

Mahdi Islam
BRAC University

Ehsanur Rahman Rhythm
University of Dhaka

Md Humaion Kabir Mehedi
BRAC University

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Sentiment Analysis of Amazon Reviews Using Machine Learning Classifier

Razin Sumyta Monsoor, Tania Sultana Tamanna, Salequzzaman Khan, Shehrin Hoque, Mahdi Islam,
Ehsanur Rahman Rhythm, Md Humaion Kabir Mehedi and Annajiat Alim Rasel
Department of Computer Science and Engineering (CSE)
School of Data and Sciences (SDS)
Brac University
66 Mohakhali, Dhaka - 1212, Bangladesh
{razin.sumyta.monsoor, tania.sultana.tamanna, salequzzaman.khan, shehrin.hoque, mahdi.islam,
ehsanur.rahman.rhythm, humaion.kabir.mehedi}@g.bracu.ac.bd, annajiat@gmail.com

***Abstract*—Reviews can significantly impact a company's reputation in the market, potentially influencing its overall business outcomes, either positively or negatively. This is especially crucial for companies that operate primarily through e-commerce platforms. Hence, it is vital for companies to pay close attention to customer reviews. Sentiment Analysis, often referred to as "opinion mining," is a significant procedure in Natural Language Processing (NLP) which serves the purpose of ascertaining the emotional tone of a provided text and categorizing it into positive, negative, or neutral perspectives. In this paper, sentiment analysis methodology is presented for classifying Amazon reviews which utilizes a large dataset of reviews and employs Multinomial Naïve Bayesian (MNB), Support Vector Machine (SVM), Maximum Entropy (ME), and Logistic Regression as the primary classifiers by the authors. With the aid of machine learning, we employed a supervised learning approach to an extensive Amazon dataset in order to categorize it based on sentiment polarity, achieving a high level of accuracy for the results. Here, we utilized the Kaggle dataset that includes a substantial volume of reviews and associated metadata which comprises customer reviews and ratings on Amazon products.**

***Index Terms*—Sentiment Analysis, Machine learning, Natural Language Processing, Naïve Bayesian (MNB), Support Vector Machine (SVM), Maximum Entropy, Logistic Regression, feature extraction, text classification.**

I. Introduction

The long-term viability of businesses like Amazon is heavily contingent on their capacity to effectively fulfill customer requirements. Countless individuals share their opinions about various services or products through different platforms like social networking sites, blogs, or popular review sites that will come up just with a Google search. So, in recent times it has become a common practice to search for reviews before making a purchase decision. Hence, disseminating customer reviews and feedback about online products or services can significantly impact the perceptions of new customers regarding the organization. Additionally, Amazon can examine these reviews to determine their authenticity and assess whether they might be part of a competitor's scheme to manipulate perceptions. Also, conducting a thorough analysis of customer sentiments empowers Amazon's business enthusiasts to gain deeper insights into the market, enabling them to make informed decisions that preemptively address customer needs and concerns, like betterment or elimination of any existing product, increasing or decreasing monetary value for marketing reach, etc. Sentiment analysis leverages natural language processing techniques and text analysis to delve into what customers are conveying, how they articulate their thoughts, and the underlying meaning behind their expressions. The primary objectives of this paper are to extract the sentiments conveyed in customer reviews and analyze these sentiments. Next, we need to develop and train a machine learning model capable of categorizing customer reviews into two sentiment categories: positive and negative. Sentiment analysis involves several key steps, including- text preprocessing, tokenization, feature extraction, sentiment classification, post-processing, and evaluation. These steps collectively contribute to the accurate analysis of sentiments in textual data. In this context, sentiment analysis relies on contextual mining of Amazon reviews to identify and extract subjective information from source materials. This process assists Amazon in gaining insights into the social sentiment surrounding their brand, products, or services while actively monitoring online conversations in a more efficient manner as they don't have the time to surf through massive databases.

Now, all of these customer reviews are huge amounts of unstructured data and sentiment analysis proves to be an efficient and cost-effective method to extract meaningful insights from these unstructured texts. The primary goal of this paper is to classify customers' positive and negative feedback on various Amazon products and construct a supervised learning model capable of categorizing a substantial volume of reviews based on sentiment. This approach utilizes data mining, machine learning (ML), and computational linguistics like Multinomial Naïve Bayesian (MNB), Support Vector Machine (SVM), Maximum Entropy (ME), and Logistic Regression to extract sentiment and subjective content from text and also assess whether the text conveys positive or negative sentiments. Through data mining, the company is able to understand the polarity of sentiment toward their respective products or services. In this paper, our model utilized both manual and active learning approaches for labeling our datasets. To achieve satisfactory results various classifiers were employed iteratively we utilized the labeled datasets to conduct further processing while the extracted features were classified through different classifiers. We utilized a combination of two feature extraction methods: the bag of words approach and the tf-idf and Chi-square approach to enhance the accuracy of our results.

II. Background

In our study "Sentiment Analysis on Amazon Reviews," we set out to explore the potential of sentiment analysis within the context of Amazon's enormous review ecosystem. We make use of a sizable dataset from Kaggle, a community for data science fans that includes a wide range of reviews and related metadata. Our objective is straightforward: to use computational linguistics and machine learning methods, such as Naïve Bayes, Multinomial Naïve Bayes, Support Vector Machine (SVM), and Decision Tree, as major classifiers to categorize these reviews based on sentiment. We seek to categorize a sizable number of reviews based on sentiment polarity by utilizing supervised learning approaches and sophisticated feature extraction techniques like the bag of words approach and TF- IDF. The goal of this research is to give organizations the resources they require to get a thorough understanding of client attitudes, empowering them to make defensible choices that satisfy customer demands, enhance current goods or services, and improve marketing tactics. In conclusion, sentiment analysis is a powerful tool that, by revealing the hidden sentiments inside customer reviews, can help firms flourish in today's cutthroat business environment. Our study goes into the complexities of sentiment analysis and offers businesses a road map for efficiently utilizing client feedback. We hope to enable businesses like Amazon to make data-driven decisions that improve customer satisfaction and propel success in the digital era through this investigation.

III. Literature Review

Many research papers regarding opinion mining or sentiment analysis have already been explored by many researchers. For identifying the interests of the customers about products, sentiment analysis played an important role at the documentation level. The authors in the work [1], developed a business model by analyzing the result of the sentiments they got from the reviews on Amazon. Also, they focused on detecting the emotions of the customers from the reviews and fake reviews. Besides, they worked on finding the gender of the customers based on the name given. Multinomial Naïve Bayesian (MNB) and Support Vector Machine(SVM) were used as the dominant classifiers in their work. From the paper [2], the author enlisted the help of a supervised machine learning algorithm for foreseeing the textual-based review ratings on a given numerical scale.

For the work, 70% of the data were used as training data and the rest 30% as testing data for cross-validation. Applying different classifiers as their algorithm, the author was determined to get accurate results of the values. The author of the paper [3], chose reviews from the section of books and Kindle on Amazon to perform sentiment analysis for research. Mainly, Naïve Bayesian (NB) and decision list classifiers were applied to determine whether to label the reviews given as positive or negative. For paper [4], the author applied Naïve Bayesian, Support Vector Machine, and Maximum Entropy as the main classifiers that created a system that portrayed the sentiment of the reviews in the chart. By using data scraping from the Amazon URL, they preprocessed the data obtained from it. In the paper, they did not show the result of accuracy as they only gave a summary of the product review. As a result, they depicted the result in the form of a statistical chart. In paper [5], the author used simple algorithms as they got high accuracy on support vector machine. However, those algorithms were incapable of performing well on a large number of datasets. So, they applied a simple algorithm of support vector machine, logistic regression, and decision tree method. The author in the paper [6], used tf-idf to speculate the given ratings from a bag of words. Besides, they did not use many classifiers. So, with the help of root mean square error and linear regression model, they got the result. Authors in paper [7], designed a model for the prediction of the ratings of the product dependent on a bag of words from rating text for testing using unigrams and bigrams. Upon testing, unigrams gave more precise results than bigrams. The percentage of unigrams was 15.89% better than the result of bigrams. In the paper [8], the authors used different types of selection techniques for doing research on sentiment analysis. Because of stop words and removing special characters, they performed preprocessing after gathering data from Amazon. Besides, they applied Naïve Bayes as the main classifier for phrase level, single word, and multiword feature selection. As a result, they summed up that the result of phrase level was better than of single word and multiword with the help of Naïve Bayes. Moreover, they only applied the algorithm of Naïve Bayes as their classifiers. The performance of Naïve Bayes was better than that of logistic regression and SentiWordNet for classifying the review as positive or negative [9]. Recall, Precision, and F-measure were used to evaluate the performance of the method used. In [10], Naïve Bayes, Support Vector Machine, Logistic Regression, and Random Forest were applied along with a continuous bag of words(CBOW) and skip-gram methods by the authors. The result obtained from the experiment displayed that Random Forest with a continuous bag of words had the most accuracy of 91% among the other algorithms. The enhanced Naïve Bayes model was built by the authors with the combination of effective negation handling, word n-grams, and feature selection methods for sentiment analysis [11]. Again here [12]the author carried out sentiment analysis on different online papers "SAOOP", where the evaluation was divided into sentiment score and system score criteria, their comparison method shows that their model results in higher accuracy than NLTK and NLP. Moreover, people nowadays also tend to show their emotions through emojis to express emotions on the internet, the investigators in reference [13] executed an assessment of emoji sentiments, analyzing emojis to derive the underlying sentiments expressed. They also provided visual tools such as a sentiment bar for better understanding of readers. In the cited paper [14], the author applied the BRDT algorithm and as well as feature extraction to furnish outcomes for sentiment analysis, they carried out their algorithm in Facebook datasets and movie datasets where the accuracy reached 50% and 99% on Facebook depending on the threshold, and 96% on movies.

As a result, we decided to use ideas from the above-mentioned works which we thought would give us higher accuracy in our research. We were able to make better decisions by making the best use of ideas from other related works. So, we were able to get enhanced results from our tasks for applying an active learning approach [15]. With the help of machine learning algorithms, the classification of textual data for sentiment analysis is done easily for the detection of positive and negative reviews about the products on Amazon. By using machine learning algorithms, we concentrated more on the preciseness of the classification of the reviews. So, we extracted data from the reviews on the Amazon site and created a business model by surveying the results of the reviews. Based on the data, we detected positive and negative reviews of the customer by using Multinomial Naïve Bayesian(MNB), Support Vector Machine(SVM), Maximum Entropy(ME), and Logistic Regression as the main classifier. By adopting a supervised machine learning algorithm, we predicted the ratings of textual reviews on a given numerical scale. In the paper, we used natural language processing for classifying the textual reviews. To label the reviews given by the customers whether it is positive or negative, Naïve Bayesian and Decision list are used as classifiers.The main objective is to create a system that portrays the sentiment of the reviews in the chart.Maximum Entropy came under the category of probabilistic classifier and it did not take into account the elements independent of each other. Also added that Maximum Entropy solved a huge amount of text classification problems.

So, it is very popular in the field of sentiment analysis. Naïve Bayesian constructed classifiers that assigned a class label to the problem examples. Depending on the value of the features being defined, Naïve Bayesian expressed a vector form where labels came from the finite sets. Despite not being a standard algorithm, Naïve Bayesian is dependent on the principles being used. For classification problems, another algorithm called Support Vector Machine is used widely. Besides, they are popular for being especially effective at separating data points that belong to various classes using the ideal hyperplane, making them particularly well-suited for classification issues. They distinguished the classes very effectively. On the other hand, Logistic Regression known as the statistical approach and machine learning algorithm on the idea of probability is used to solve classification problems. It is applicable if the dependent variable is categorical. To determine whether the output of the reviews is positive, neutral, or negative, we used logistic regression. As a result, to get more accurate results for our research paper, we used logistic regression, Support Vector Machine, Naïve Bayesian, and Maximum Entropy for the prediction of Amazon reviews.

IV. Methodology

A. Collected Data

In our effort to conduct sentiment analysis on Amazon reviews, we made use of a carefully curated dataset obtained from Kaggle, a well-known community for data science enthusiasts. We used the Kaggle dataset for our investigation, which contains a sizable number of reviews and related metadata, to build a strong sentiment analysis framework. 21 columns and 34,660 rows make up this dataset, which depicts various facets of the review data in each row and column. We attempted to enhance result accuracy by consolidating three distinct Excel sheets from the provided dataset, resulting in an aggregate of over 65 thousand rows. We concentrated on using 10 essential features from this large dataset for our particular investigation. Our sentiment analysis targets extracting valuable insights from the Kaggle dataset, illuminating the sentiments expressed within Amazon reviews by focusing on these 10 carefully selected features. This group of features provides a targeted and effective way to analyze consumer sentiment on Amazon and discover the variables affecting how they feel about specific products. During the coding process to evaluate the precision and accuracy of various algorithms, our emphasis was on the '*review.rating*' and '*review.comments*' aspects.

B. Proposed Methodology

One of Amazon's most valuable and established features is customer reviews. For the purpose of determining the polarity of a review, we have used the *reviews.text* and *reviews.rating* features from our dataset. We have preprocessed our data by removing null values from our data and replacing them with spaces, as well as any extraneous features, since If they are not handled properly, missing values could introduce bias and errors into the analysis. Missing values have the potential to skew statistical metrics like means, medians, and correlations. There are a finite number of categorical variables, most of which take the form of "strings" or "categories". Data preparation is made more challenging by categorical values' lack of numeric values. Algorithms can find relationships and patterns in the data by converting category values into numerical representations, which are widely used to describe qualitative aspects. Additionally, categorical variables like reviews and text are included in our project. The code in our project converts the *reviews.text* column from a string to an integer type. CountVectorizer and TfidfTransformer are used to convert the text data (reviews) into numerical characteristics that machine learning models can use. The dataset was separated into training and training datasets 70 and 30%, respectively, and scrambled at random. The accuracy was then assessed using machine learning approaches such as SVM, logistic regression, Decision tree, Naïve Bayes, and random forest algorithm.

C. Data Analysis

Data analysis is a vital phase in any machine learning project, especially one focusing on sentiment analysis from Amazon reviews, where data quality and preprocessing play a critical role. In this portion of the paper, we present a complete examination of the dataset utilized in this work, including its characteristics, preprocessing steps, model training, and testing, and come up with strategies to attain our goal. Two significant concerns such as handling missing values and handling categorical variables have stood out in our data analysis process and have ramifications on the quality and dependability of our conclusions. Missing values within our dataset, particularly in the *reviews.rating* column, provide a considerable issue. These null values can bring biases and mistakes into our analysis, influencing statistical measurements like means and correlations. Moreover, during model training and testing, these missing variables might lead to errors, hurting the performance of the machine learning models. To overcome this issue, we apply approaches like deleting null entries or imputing them with acceptable values, ensuring the integrity of our dataset. Categorical variables, such as the *reviews.text*, are non-numeric in nature, complicating data preparation. However, translating these category values into numerical representations is necessary for machine learning algorithms to recognize patterns and relationships effectively. We address this difficulty by applying text vectorization algorithms, including CountVectorizer and TfidfTransformer, which turn textual data into numerical representations suited for analysis. Moving beyond data preprocessing, we tested the performance of various machine learning models. Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), and Decision Tree classifiers were applied for model training and testing. Our results reveal that the SVM and logistic regression models regularly outperform others, obtaining a remarkable accuracy rate of roughly 93%. These models have proved their effectiveness in capturing sentiment trends among Amazon reviews. In summary, our data analysis highlights the significance of robust data pretreatment to handle missing values and categorical categories effectively. Furthermore, our model evaluation reveals both SVM and logistic regression classifiers are well-suited for our sentiment analysis project, continuously generating excellent accuracy rates. These findings give a solid platform for making informed decisions based on Amazon reviews and deriving important insights from customer sentiment.

V. Prototype Implementation

We incorporated a variety of machine-learning methods for sentiment analysis into our prototype implementation.

A. Multinomial Naïve Bayes

For our paper, we imported Naïve Bayes and used its MultinomialNB classifier with specified parameters to handle unseen features, binary predictions were made. Multinomial Naïve Bayes (MNB) is widely employed for text data with discrete features like word frequency counts. It starts by splitting the dataset into training and testing subsets. The accuracy is then generated through the confusion matrix heap. We assessed the model's performance by creating a classification report with precision and recall metrics and evaluating prediction accuracy. This algorithm is effective when dealing with discrete data, such as frequency counts. It is commonly utilized in natural language processing scenarios, like here analyzing sentiments behind the comments.

B. Support Vector Machine (SVM)

Support Vector Machine SVM is utilized for regression and classification. It finds a hyperplane that maximizes the margin between the data while simultaneously effectively classifying the data into different groups. For this method's initial step, we used the sci-kit-learn module to build a pipeline for text classification. This pipeline included SVM classification, TF-IDF transformation, and Bag-of-Words transformation. We used this pipeline to make predictions on the test data after training it on the training set of data. Finally, we calculated and displayed the SVM classifier's prediction accuracy.

C. Logistic Regression

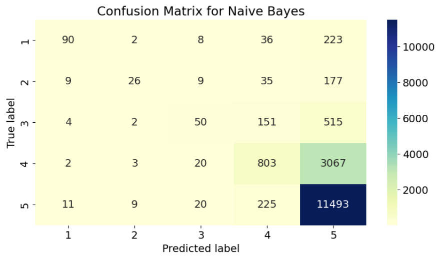
Logistic regression is a statistical method used to predict the likelihood of an event happening. Logistic regression usually encompasses three distinct models: Binary, Multinomial, and Ordinal logistic regression. Logistic regression falls under the category of discriminative models, aiming to differentiate between classes. Logistic regression is frequently employed in addressing prediction, detection, and classification challenges. Our logistic regression model rounds the predicted values to convert them into binary predictions. Then, it calculates and prints the accuracy and classification report using the true labels and the predicted labels, then it creates a data frame from the confusion matrix for better visualization. The visualization provides insights into how well the model is performing for each class.

D. Decision Tree

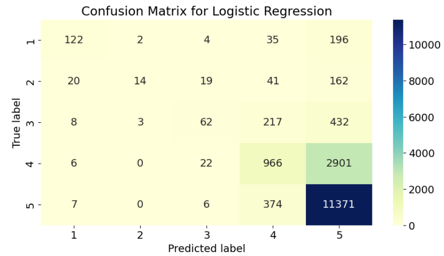
A Decision Tree is a supervised learning method that can be applied to problems related to regression and classification. It is a tree-structured classifier, where internal nodes stand in for the dataset's features, branches for the rules of classification, and each leaf node for the result. We used tools like sci-kit-learn in our implementation and built a decision tree classifier. This classifier was created to make predictions using features extracted from the training set of data. Recursively dividing the data into subsets based on particular feature criteria allowed the decision tree to learn to make decisions, eventually forming a tree-like structure. For our analysis, the tree's depth and structure were set up as needed. The decision tree classifier was used to predict sentiment labels for the test data after it had been trained.

E. Random Forest

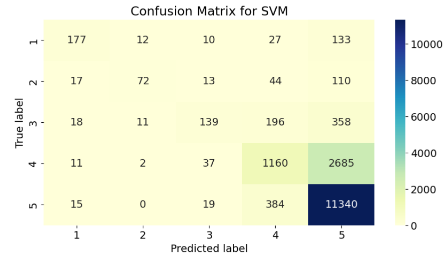
The random forest algorithm extends the bagging method by incorporating bagging and generating random features, forming different decision trees ensuring they are not related to each other. This algorithm comprises decision trees, each built from a bootstrap sample, which is a data set replacement from the training set. Our Random Forest classifier evaluates



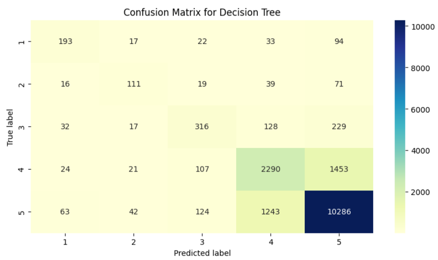
(a) Multinomial Naïve Bayes



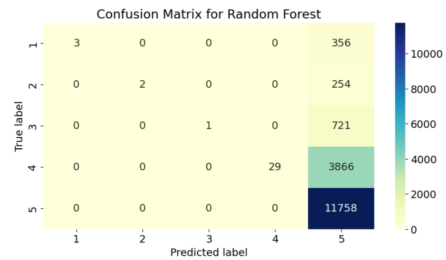
(b) Logistic Regression



(c) SVM



(d) Decision Tree



(e) Random Forest

Fig. 1: Confusion Matrix for the Machine Learning Models

its accuracy, and then makes predictions on the test set. After printing additional metrics, it computes and visualizes the confusion matrix using Seaborn. The confusion matrix provides insights into the model's performance across different classes.

VI. Result Analysis

We used Tf-idf and countVectorization for counting the polarity and relative frequency of a word. Text preprocessing, SVM classifier, Multinomial NB, Decision Tree, Logistic Regression, and Natural Language Matplotlib, are the ideas and models that are used in this article. All of these techniques have been combined in a novel way in our research to examine consumer re-perspectives where we evaluate how accurate the models we used are. We identify the review's attitude as either positive, neutral, or negative based on the comparison of the applied models' accuracy levels. Our results show that the research's outcomes are more pleasing. With the help of three matrices—precision, recall, and f1-score — we develop a classification report to evaluate the model's quality.1. The percentage of correctly made positive predictions compared to all positive predictions is known as precision. The percentage of correctly predicted positive outcomes compared to all actual positive outcomes is known as recall. The precision and recall components of the F1-Score are weighted harmonic means. Better models are those that are closest to 1. F1-Score equation is denoted in equation 1.

$$2 * (Precision * Recall)/(Precision + Recall) \quad (1)$$

By evaluating a classification model's performance using these three criteria, we may determine how effectively it can forecast results for a certain response variable.

TABLE I: Evaluation Metrics of the Machine Learning Models

Models	Precision	Recall	F1-Score	Accuracy
Multinomial Naïve Bayes	0.65	0.32	0.37	0.73
SVM	0.71	0.45	0.51	0.76
Logistic Regression	0.68	0.34	0.38	0.74
Decision Tree	0.62	0.57	0.60	0.78
Random Forest	0.94	0.20	0.17	0.69

The Naïve Bayes theorem states that the accuracy rate is 73%. In comparison, the accuracy rate for SVM is almost identical to that of logistic regression at 76%. The decision tree classifier also has a 78% accuracy rate. Therefore, we can say that Decision Tree, SVM, and Logistic Regression models are the best for our project, whereas the Random Forest model is the worst.

VII. Conclusion

Our research paper's main emphasis lies in converting the diverse range of sentiments expressed in customer reviews of Amazon products into either positive or negative sentiments. We employed sentiment analysis within the field of Natural Language Processing (NLP), which assists in identifying the emotional tone within a given text and categorizing it into positive or negative perspectives. Assembling a sizable data set posed challenges because e-commerce websites typically impose restrictions on making their data publicly available, especially when it's a big site like Amazon. To enhance accuracy and precision, we consolidated three independently provided data sets containing reviews and ratings into a single data set.

Additionally, real-time reviews further complicate data scraping efforts. For the classification of the customer reviews data set into different sentiment groups, we employed a Naïve Bayes and decision list as a classification model, which demonstrated strong performance. Apart from that we also implemented Support Vector Machine, Maximum Entropy (ME), and Logistic Regression as classifiers. Our proposed model combines two types of extraction methods. We managed to attain an accuracy rate exceeding 86%, with the F1 measure, while precision and recall also surpassed 90%. Evaluating human sentiments with a high degree of precision is a demanding endeavor. Different studies are being conducted where various models and procedures are being used to bring out higher accuracy rates. As part of our research efforts, we conducted a comprehensive review of various research papers focused on sentiment analysis and text classification techniques. This review likely provided valuable insights and informed the development of our methodology.

FUTURE WORK

In the future, we can explore the possibility of incorporating new analytical methods to further enhance the accuracy rates of our sentiment analysis. We could incorporate artificial intelligence to identify different parts of speech as features. We could also use Convolutional Neural Network (CNN) as a classifier which is a deep learning method and for extraction we could try using the K-mean clustering algorithm which is said to be effective for larger datasets. We also plan on incorporating (the PCA) principal component analysis model into our research which could make data automation a lot easier. This methodology could help in dimensional reduction while also providing recommendations to customers.

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